

AI-Assisted Literature Review, Research Gap Analysis and Publication-Oriented Review Paper Development

Prompt Engineering for Generative AI - Case Study 2

Final Review Topic: Retrieval-Augmented Generation for Hallucination Mitigation in Large Language Models: A Systematic Review of Architectures, Evaluation and Open Problems

Student Information

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Provenance of this submission

All 45 bibliographic records were resolved programmatically against the CrossRef and arXiv APIs by the submitted `fetch_papers.py` - titles, authors, years and venues come from publisher records, not from a language model and not transcribed by hand. No citation in this submission was supplied by AI. Numeric results quoted from papers carry a confidence flag in `comparison.csv`; rows marked 'check' are listed in Appendix B and must be re-read against the source before final submission.

Phase 1 - Research Domain Identification

Research Domain Selection

Domain	Selected
Generative AI	X
Machine Learning	
Deep Learning	
Computer Vision	
Cybersecurity	
Cloud Computing	
IoT	
FPGA	
VLSI	
Embedded Systems	
Data Analytics	
Other	

Topic Narrowing

Level	Topic
Broad Domain	Artificial Intelligence
Sub Domain	Generative AI / Large Language Models
Research Area	Retrieval-Augmented Generation (RAG)
Specific Focus	Hallucination mitigation and grounding verification in RAG pipelines
Final Review Topic	Retrieval-Augmented Generation for Hallucination Mitigation in Large Language Models: A Systematic Review of Architectures, Evaluation and Open Problems

Q1. Why did you select this topic?

I chose this topic because it is the direct continuation of my previous case study, where I measured how often an unconstrained language model asserts things it cannot support. That study established the problem empirically on a single topic: 37.5% of the specific factual statements the model volunteered could not be traced to any source. What it could not tell me was whether the research community has a principled solution. Retrieval-Augmented Generation is the most widely deployed answer to that question, so reviewing it systematically is the natural next step.

The topic is also well suited to a literature review rather than an implementation project. It has a clear origin point in 2020, a dense and well-indexed body of work, several competing architectural families, and an active disagreement about whether the problem is solvable at all. That disagreement matters: a review can contribute something by mapping it, whereas a topic where everyone agrees would only need summarising.

Q2. What industrial relevance does this topic have?

Retrieval-Augmented Generation is now the default architecture for deploying language models against private or current data. Enterprise search, customer support automation, clinical decision support, legal document review and internal knowledge assistants are all built on it, because the alternative - fine-tuning a model whenever the underlying data changes - is economically impossible for corpora that change daily.

The industrial stake is specifically about liability rather than accuracy in the abstract. A system that is usually right but occasionally fabricates a citation, a dosage or a contractual clause cannot be deployed without human review, and the cost of that review often exceeds the saving from automation. This is visible in the literature itself: [P34] evaluates twelve RAG variants for clinical decision support and treats on-premises deployment and provenance tagging as first-class requirements, and [P35] re-

ports that residual hallucination across three industrial deployments still required human oversight. Regulation is moving the same way, since obligations to explain automated decisions are difficult to satisfy when a system cannot say where its answer came from. Retrieval provides exactly that provenance, which is why it is an architectural requirement and not only a quality improvement.

Q3. Why is this topic currently important?

Three things converged to make this urgent between 2023 and 2026.

First, capability outran verifiability. Models became fluent enough that their output stopped carrying surface cues of unreliability, so readers lost the informal signal they had been using to calibrate trust.

Second, the theoretical picture changed. [P15] argues formally that hallucination is an innate limitation and cannot be eliminated, and [P16] argues that prevailing evaluation practice actively rewards guessing over admitting ignorance. Together these reframed the engineering objective from elimination to bounded, measurable residual risk - a substantially different target.

Third, the evidence base matured. Until 2024 there was no large corpus of word-level hallucination annotations over genuinely retrieval-augmented output; [P32] supplied one, and it showed something uncomfortable: RAG systems still hallucinate even when the correct evidence has been successfully retrieved. That finding undercuts the central assumption of the paradigm and is the single most important reason this review is worth writing now.

Q4. What challenges exist in this area?

Four challenges shaped how this review had to be conducted.

The first is that numeric results are not comparable across papers. Reported systems use different base models, retrievers, corpora and splits, so a leaderboard assembled from published numbers ranks experimental setups rather than methods. I treat this as a methodological constraint throughout and report architectural comparison instead, stating explicitly where numeric comparison would be invalid.

The second is terminological drift. "Hallucination" means intrinsic unfaithfulness to a source in [P12], factuality error against world knowledge in [P13], and unsupported spans relative to retrieved context in [P32]. A review that does not fix its definition ends up comparing measurements of different phenomena.

The third is the pace of publication. The field produces preprints faster than peer review can absorb them, so a large share of the most-cited work is unrefereed. I include preprints where they are foundational or genuinely current, but record publication status so the reader can weight them.

The fourth is evaluation circularity. Several standard metrics use a language model as the judge. If judge and generator share a misconception, the metric certifies the error rather than catching it. This is a live threat to the validity of the numbers this review reports, and it became one of the gaps.

Phase 2 - Research Paper Discovery

45 papers collected - 35 carried through the full Phase 3 matrix and 10 additional verified works cited for context. Requirement is a minimum of 30.

Publisher family	Papers
arXiv (preprint)	16
ACL Anthology	15
IEEE	3
ACM	3
MIT Press (TACL)	2
Springer / Nature	2
MDPI	2
Elsevier	1
Wiley	1

Year	2019	2020	2021	2022	2023	2024	2025	2026
Papers	1	3	3	2	11	14	7	4

Paper Repository Master Sheet

Sr	ID	Paper Title	Author(s)	Year	Publisher / Venue	Identifier	Rel.
1	P01	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	Lewis et al.	2020	Adv. Neural Inf. Process. Syst. (NeurIPS)	arXiv:2005.11401	5
2	P02	Dense Passage Retrieval for Open-Domain Question Answering	Karpukhin et al.	2020	Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)	10.18653/v1/2020.emnlp-main.550	5
3	P03	REALM: Retrieval-Augmented Language Model Pre-Training	Guu et al.	2020	Proc. Int. Conf. Mach. Learn. (ICML)	arXiv:2002.08909	4
4	P04	Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering	Izacard et al.	2021	Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume	10.18653/v1/2021.eacl-main.74	5
5	P05	Improving language models by retrieving from trillions of tokens	Borgeaud et al.	2022	Proc. Int. Conf. Mach. Learn. (ICML)	arXiv:2112.04426	4
6	P06	Billion-Scale Similarity Search with GPUs	Johnson et al.	2021	IEEE Transactions on Big Data	10.1109/tbdata.2019.2921572	4
7	P07	In-Context Retrieval-Augmented Language Models	Ram et al.	2023	Transactions of the Association for Computational Linguistics	10.1162/tacl_a_00605	4
8	P08	REPLUG: Retrieval-Augmented Black-Box Language Models	Shi et al.	2024	Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)	10.18653/v1/2024.naacl-long.463	4
9	P09	Language Models as Knowledge Bases?	Petroni et al.	2019	Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)	10.18653/v1/d19-1250	4
10	P10	When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories	Mallen et al.	2023	Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)	10.18653/v1/2023.acl-long.546	5
11	P11	Lost in the Middle: How Language Models Use Long Contexts	Liu et al.	2024	Transactions of the Association for Computational Linguistics	10.1162/tacl_a_00638	5
12	P12	Survey of Hallucination in Natural Language Generation	Ji et al.	2023	ACM Computing Surveys	10.1145/3571730	5
13	P13	A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions	Huang et al.	2025	ACM Transactions on Information Systems	10.1145/3703155	5

Sr	ID	Paper Title	Author(s)	Year	Publisher / Venue	Identifier	Rel.
14	P14	Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models	Zhang et al.	2023	arXiv preprint	arXiv:2309.01219	4
15	P15	Hallucination is Inevitable: An Innate Limitation of Large Language Models	Xu et al.	2024	arXiv preprint	arXiv:2401.11817	4
16	P16	Why Language Models Hallucinate	Kalai et al.	2025	arXiv preprint	arXiv:2509.04664	4
17	P17	SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models	Manakul et al.	2023	Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing	10.18653/v1/2023.emnlp-main.557	5
18	P18	Detecting hallucinations in large language models using semantic entropy	Farquhar et al.	2024	Nature	10.1038/s41586-024-07421-0	5
19	P19	TruthfulQA: Measuring How Models Mimic Human Falsehoods	Lin et al.	2022	Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)	10.18653/v1/2022.acl-long.229	4
20	P20	FACTScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation	Min et al.	2023	Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing	10.18653/v1/2023.emnlp-main.741	5
21	P21	AutoHall: Automated Factuality Hallucination Dataset Generation for Large Language Models	Cao et al.	2026	IEEE Transactions on Audio, Speech and Language Processing	10.1109/taslp.2025.3635038	4
22	P22	Retrieval Augmentation Reduces Hallucination in Conversation	Shuster et al.	2021	Findings of the Association for Computational Linguistics: EMNLP 2021	10.18653/v1/2021.findings-emnlp.320	5
23	P23	Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection	Asai et al.	2024	Proc. Int. Conf. Learn. Represent. (ICLR)	arXiv:2310.11511	5
24	P24	Corrective Retrieval Augmented Generation	Yan et al.	2024	arXiv preprint	arXiv:2401.15884	5
25	P25	Active Retrieval Augmented Generation	Jiang et al.	2023	Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing	10.18653/v1/2023.emnlp-main.495	5
26	P26	Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions	Trivedi et al.	2023	Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)	10.18653/v1/2023.acl-long.557	5
27	P27	Chain-of-Verification Reduces Hallucination in Large Language Models	Dhuliawala et al.	2024	Findings of the Association for Computational Linguistics ACL 2024	10.18653/v1/2024.findings-acl.212	5
28	P28	RARR: Researching and Revising What Language Models Say, Using Language Models	Gao et al.	2023	Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)	10.18653/v1/2023.acl-long.910	4
29	P29	From Local to Global: A Graph RAG Approach to Query-Focused Summarization	Edge et al.	2024	arXiv preprint	arXiv:2404.16130	5
30	P30	RAGAs: Automated Evaluation of Retrieval Augmented Generation	Es et al.	2024	Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations	10.18653/v1/2024.eacl-demo.16	5
31	P31	Benchmarking Large Language Models in Retrieval-Augmented Generation	Chen et al.	2024	Proc. AAAI Conf. Artif. Intell.	arXiv:2309.01431	5
32	P32	RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models	Niu et al.	2024	Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)	10.18653/v1/2024.acl-long.585	5
33	P33	Hallucination Mitigation for Retrieval-Augmented Large Language Models: A Review	Zhang et al.	2025	Mathematics	10.3390/math13050856	5
34	P34	Evaluating Retrieval-Augmented Generation Variants for Clinical Decision Support: Hallucination Mitigation and Secure On-Premises Deployment	Wolk	2025	Electronics	10.3390/electronics14214227	5
35	P35	Large Language Model Hallucination Mitigation in Three Industrial Use Cases	Tikka et al.	2026	IEEE Access	10.1109/access.2026.3659997	4
36	R36	Retrieval-Augmented Generation for Large Language Models: A Survey	Gao et al.	2023	arXiv preprint	arXiv:2312.10997	4
37	R37	A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models	Fan et al.	2024	Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining	10.1145/3637528.3671470	4
38	R38	Large language models hallucination: A comprehensive survey	Alansari et al.	2026	Computer Science Review	10.1016/j.cosrev.2026.100970	4
39	R39	Hallucination to truth: a review of fact-checking and factuality evaluation in large language models	Rahman et al.	2026	Artificial Intelligence Review	10.1007/s10462-025-11454-w	4

Sr	ID	Paper Title	Author(s)	Year	Publisher / Venue	Identifier	Rel.
40	R40	Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG	Singh et al.	2025	arXiv preprint	arXiv:2501.09136	4
41	R41	Graph Retrieval-Augmented Generation: A Survey	Peng et al.	2024	arXiv preprint	arXiv:2408.08921	4
42	R42	RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval	Sarathi et al.	2024	Proc. Int. Conf. Learn. Represent. (ICLR)	arXiv:2401.18059	4
43	R43	A Stitch in Time Saves Nine: Detecting and Mitigating Hallucinations of LLMs by Validating Low-Confidence Generation	Varshney et al.	2023	arXiv preprint	arXiv:2307.03987	4
44	R44	MedVH: Toward Systematic Evaluation of Hallucination for Large Vision Language Models in the Medical Context	Gu et al.	2025	Advanced Intelligent Systems	10.1002/aisy.202500255	3
45	R45	Mitigating Hallucination in Large Language Models (LLMs): An Application-Oriented Survey on RAG, Reasoning, and Agentic Systems	Li et al.	2025	arXiv preprint	arXiv:2510.24476	4

How did you identify these papers?

Three passes. First, a seed set of architecture-defining papers ([P01], [P02], [P03]) identified from recent surveys ([R36], [P33]). Second, forward and backward citation chaining from that seed set to find both the antecedents and the methods that build on them. Third, systematic database queries against CrossRef and the arXiv API, filtered to 2015-present and to IEEE, Springer, ACM, Elsevier, MDPI, Wiley and arXiv, to catch peer-reviewed journal work that citation-chaining from preprints tends to miss. Every record was then resolved against the publisher's own metadata (see Phase 2 methodology note) rather than transcribed from a citation.

What keywords did you use?

Core: "retrieval-augmented generation", "RAG", "hallucination mitigation", "factuality", "grounding", "attribution". Architecture: "dense retrieval", "adaptive retrieval", "self-reflective RAG", "corrective RAG", "graph RAG", "iterative retrieval". Evaluation: "hallucination detection", "faithfulness metric", "RAG benchmark", "factual precision". Boolean pairs such as ("retrieval-augmented" AND "hallucination") were the most productive; "RAG" alone returns heavy noise from unrelated fields, which is why it was never used unqualified.

How did AI assist in paper discovery?

AI was used for three things and deliberately not for a fourth. It expanded the keyword set by proposing synonyms and adjacent terminology I had not considered ('grounding', 'attribution', 'faithfulness' as distinct from 'factuality'). It clustered the collected set into thematic groups, which produced the seven-theme structure used in Phase 3. It drafted per-paper summaries against a fixed schema. It was NOT used to supply citations: every attempt to have a model produce bibliographic details directly yielded at least one plausible but non-existent record, consistent with the fabricated-citation failure mode measured in my previous case study. All 45 records here were resolved programmatically against CrossRef and the arXiv API, and the resolution script is submitted with this report.

Phase 3 - Literature Review Master Matrix

All 35 primary studies analysed against a fixed eight-field schema. Fields not stated in a paper were recorded as such rather than inferred.

Paper	Research Problem	Methodology	Dataset / Benchmark	Evaluation Metrics
P01 - Lewis et al. (2020)	Parametric LLMs cannot cite or update the knowledge they encode, so factual errors are untraceable and unfixable without retraining.	Fine-tuned generator (BART) conditioned on passages from a dense retriever (DPR) over a Wikipedia index; RAG-Sequence and RAG-Token variants marginalise over retrieved documents.	Natural Questions, TriviaQA, WebQuestions, CuratedTrec, MS MARCO, FEVER; Wikipedia dump as index.	Exact Match, F1, generation quality, factual correctness of generations.
P02 - Karpukhin et al. (2020)	Sparse lexical retrieval (BM25) misses semantically relevant passages that share no vocabulary with the query.	Dual-encoder trained with in-batch negatives and hard negatives to embed questions and passages into a shared dense space; retrieval by maximum inner product search.	Natural Questions, TriviaQA, WebQuestions, CuratedTREC, SQuAD; Wikipedia passage index.	Top-k retrieval accuracy, downstream QA Exact Match.
P03 - Guu et al. (2020)	Language model pre-training stores knowledge implicitly in weights, which is opaque and cannot be inspected or edited.	Latent knowledge retriever trained jointly with masked language model pre-training, with retrieval treated as a latent variable and back-propagated through.	Wikipedia pre-training corpus; evaluated on Natural Questions, WebQuestions, CuratedTrec.	Exact Match on open-domain QA.
P04 - Izacard et al. (2021)	Extractive readers cannot synthesise an answer that requires combining several retrieved passages.	Fusion-in-Decoder: encode each retrieved passage independently with the encoder, then let the decoder attend jointly over all passage representations.	Natural Questions, TriviaQA.	Exact Match.
P05 - Borgeaud et al. (2022)	Scaling parametric knowledge by growing model size is extremely expensive.	Retrieval from a trillion-token database using frozen BERT embeddings and chunked cross-attention into a transformer decoder.	MassiveText; evaluated on language modelling and downstream tasks; Pile and Wikitext103.	Perplexity (bits per byte), downstream task accuracy.
P06 - Johnson et al. (2021)	Similarity search over billions of dense vectors is infeasible with exact methods on CPU.	GPU-optimised approximate nearest neighbour search with product quantisation and an efficient k-selection algorithm.	SIFT1B, DEEP1B benchmark vector collections.	Recall@k against exact search, queries per second, index build time.
P07 - Ram et al. (2023)	Retrieval augmentation usually requires modifying or fine-tuning the language model, which is impossible for closed models.	Prepend retrieved documents to the input context of an off-the-shelf frozen LM; study retriever choice and retrieval stride.	WikiText-103, The Pile subsets, and open-domain QA sets.	Perplexity, downstream QA accuracy.
P08 - Shi et al. (2024)	The strongest LLMs are only available behind APIs, so retrieval cannot be integrated into their internals.	Treat the LM as a black box; prepend retrieved documents and train the retriever using LM likelihood as the supervision signal (REPLUG LSR).	Pile, Natural Questions, TriviaQA, MMLU.	Perplexity, Exact Match, accuracy.
P09 - Petroni et al. (2019)	It is unclear how much relational factual knowledge language models actually store and can reliably recall.	LAMA probe: cloze-style queries derived from knowledge bases issued to pre-trained LMs without fine-tuning.	Google-RE, T-REx, ConceptNet, SQuAD-derived probes.	Precision@k (mean precision at 1 and 10).
P10 - Mallen et al. (2023)	Retrieval is applied indiscriminately even where the model already knows the answer, which adds cost and can hurt accuracy.	Construct PopQA around entity popularity; measure parametric accuracy against subject-entity popularity; compare with non-parametric retrieval; propose adaptive retrieval by popularity threshold.	PopQA (constructed), EntityQuestions, Natural Questions.	Accuracy stratified by entity popularity.
P11 - Liu et al. (2024)	It is assumed that adding more retrieved context monotonically helps; the position of evidence in the context may matter.	Controlled experiments on multi-document QA and key-value retrieval, varying the position of the relevant document within the input context.	Multi-document QA built from Natural Questions; synthetic key-value retrieval task.	Accuracy as a function of gold-evidence position and context length.
P12 - Ji et al. (2023)	Hallucination is used loosely across NLG subfields with no shared definition, taxonomy or metric.	Systematic survey; defines intrinsic versus extrinsic hallucination; organises causes, metrics and mitigation across summarisation, dialogue, translation and QA.	Survey of task-specific benchmarks across NLG.	Reviews statistical and model-based faithfulness metrics.

Paper	Key Findings	Advantages	Limitations	Future Scope
P01 - Lewis et al. (2020)	Retrieval-augmented generation outperforms purely parametric seq2seq on open-domain QA and produces more specific and factual language; knowledge can be swapped by replacing the index without retraining.	Establishes the end-to-end RAG paradigm; hot-swappable knowledge; provenance is available because retrieved passages are explicit.	Retriever is largely frozen and errors propagate; single-shot retrieval cannot serve multi-hop questions; no mechanism to detect when the retrieved evidence fails to support the output.	Joint retriever-generator training, multi-step retrieval, and explicit verification of generated claims against retrieved evidence.
P02 - Karpukhin et al. (2020)	Dense retrieval trained on relatively few question-passage pairs substantially outperforms BM25 on top-20 retrieval accuracy, and the gain carries through to end-task QA.	Became the default retriever for RAG systems; simple, efficient, trainable with modest supervision.	Requires supervised question-passage pairs; degrades out of domain; embeddings are static so the index must be rebuilt when the encoder changes.	Unsupervised and domain-adaptive dense retrieval; hybrid sparse-dense fusion.
P03 - Guu et al. (2020)	Pre-training the retriever jointly with the language model yields large gains over models that treat retrieval as a fixed pre-processing step, and makes the knowledge source interpretable.	First to train retrieval end-to-end during pre-training; knowledge is modular and inspectable.	Pre-training is computationally very expensive; asynchronous index refresh is complex; evaluated only on extractive-style QA.	Cheaper joint training and extension beyond short-answer QA.
P04 - Izacard et al. (2021)	Performance scales strongly with the number of retrieved passages, showing that generative readers aggregate evidence across passages rather than selecting one.	Simple, highly effective, and scales with retrieved context; long the strongest open-domain QA reader.	Cost grows linearly with passage count; no notion of evidence sufficiency; cannot flag unsupported output.	Efficient attention over many passages; evidence selection and confidence estimation.
P05 - Borgeaud et al. (2022)	A model with far fewer parameters matches much larger parametric models when given retrieval over a very large corpus, indicating knowledge can be off-loaded from weights to an index.	Demonstrates a favourable compute-knowledge trade-off at scale; retrieval at unprecedented database size.	Frozen retriever; huge engineering cost for the index; potential test-set leakage from the retrieval corpus requires careful deduplication.	Trainable large-scale retrievers and rigorous leakage control.
P06 - Johnson et al. (2021)	Billion-scale k-NN search becomes practical on a small number of GPUs with orders-of-magnitude speedups over prior CPU systems at comparable recall.	Infrastructure that makes production RAG feasible; widely used as the FAISS library.	Approximation reduces recall, and that recall loss silently becomes a retrieval failure upstream of the generator; tuning is workload-specific.	Hardware-aware indexing and better accuracy-latency trade-offs.

Paper	Key Findings	Advantages	Limitations	Future Scope
P07 - Ram et al. (2023)	Simply placing retrieved text in the prompt of a frozen LM gives substantial language-modelling gains, and the retriever can be tuned to the LM without touching the LM.	No model access required; immediately deployable; clean baseline for prompt-level RAG.	Bounded by context window; irrelevant retrieved text actively harms output; no filtering of unhelpful passages.	Learned reranking and adaptive decisions about when to retrieve.
P08 - Shi et al. (2024)	Tuning the retriever against black-box LM feedback improves performance on language modelling and knowledge-intensive tasks without any model access.	Practical for commercial APIs; the retriever adapts to the specific consumer model.	Requires token-likelihood access, which many APIs no longer expose; still single-shot retrieval.	Retriever adaptation from coarser feedback signals such as preferences or ratings.
P09 - Petroni et al. (2019)	LMs recall a surprising amount of relational knowledge without supervision, but recall is uneven and highly sensitive to how the query is phrased.	Foundational evidence that parametric memory is real but unreliable - the empirical motivation for retrieval augmentation.	Cloze probes measure recall of frequent relations only; prompt sensitivity confounds the measurement of what is actually 'known'.	Better probing methodology and separating knowledge from surface-form sensitivity.
P10 - Mallen et al. (2023)	Parametric memory works well for popular entities and fails sharply on long-tail entities; retrieval helps most on the tail and can actively degrade answers on head entities.	Provides the empirical case for *adaptive* rather than always-on retrieval; cheap popularity heuristic works well.	Popularity is a crude proxy for model confidence; results centred on entity-centric factoid QA.	Confidence-aware and self-knowledge-driven retrieval triggers.
P11 - Liu et al. (2024)	Performance is highest when the relevant evidence is at the beginning or end of the context and degrades markedly when it sits in the middle - a U-shaped curve robust across models.	Explains a large class of RAG failures that look like retrieval successes; directly motivates reranking and compression.	Primarily on the model generation of its time; mechanism behind the effect is not established.	Position-robust architectures, evidence reranking, and context compression.
P12 - Ji et al. (2023)	Hallucination arises from both data (source-reference divergence) and modelling/inference; no single metric captures it, and task-specific definitions differ substantially.	The standard reference taxonomy; the intrinsic/extrinsic split is used throughout the field.	Predates instruction-tuned LLMs and RAG-specific failure modes; largely encoder-decoder framing.	Unified evaluation and hallucination analysis for large-scale generative models.

Paper	Research Problem	Methodology	Dataset / Benchmark	Evaluation Metrics
P13 - Huang et al. (2025)	LLM hallucination needs a taxonomy and consolidated account of causes, detection and mitigation specific to modern LLMs.	Large-scale survey organised around factuality and faithfulness hallucination, with causes traced to data, training and inference stages.	Reviews the major LLM hallucination benchmarks.	Surveys factuality metrics, benchmarks and human evaluation protocols.
P14 - Zhang et al. (2023)	Hallucination research is fragmented across detection, benchmarks and mitigation with inconsistent terminology.	Survey organising hallucination in LLMs into evaluation, explanation and mitigation, with a taxonomy of sources.	Reviews existing hallucination benchmarks.	Surveys automatic and human factuality evaluation.
P15 - Xu et al. (2024)	Whether hallucination can be eliminated at all, or is a structural property of the model class.	Formal argument using a computability framing: defines a ground-truth function and shows LLMs cannot learn all such functions, so hallucination is unavoidable in general.	Theoretical; no empirical dataset.	Not applicable - formal result.
P16 - Kalai et al. (2025)	Why models produce confident falsehoods rather than abstaining.	Analysis of pre-training and post-training objectives, arguing hallucination is the statistically optimal response under binary scoring that penalises abstention.	Analytical, with reference to standard benchmark scoring practice.	Discusses benchmark scoring schemes and their incentives.
P17 - Manakul et al. (2023)	Detecting hallucination usually requires an external knowledge base or access to model internals, neither of which is available for black-box APIs.	Sample multiple stochastic generations and measure consistency between them; low agreement across samples signals fabrication. Variants use BERTScore, QA-based and n-gram consistency.	WikiBio-derived hallucination annotations with sentence-level labels.	AUC-PR for sentence-level detection, correlation with human factuality judgements.
P18 - Farquhar et al. (2024)	Distinguishing uncertainty about meaning from uncertainty about wording, in order to detect confabulation reliably.	Semantic entropy: cluster sampled generations into semantic equivalence classes using bidirectional entailment and compute entropy over meanings rather than token sequences.	Open-domain QA and biography generation across several model families.	AUROC for confabulation detection, comparison with token-level entropy baselines.
P19 - Lin et al. (2022)	Models may reproduce popular human misconceptions, which standard accuracy benchmarks do not capture.	Adversarially constructed questions where a common human misconception is the tempting answer; human and automated (GPT-judge) evaluation.	TruthfulQA: 817 questions across 38 categories.	Percentage of truthful answers, truthful-and-informative percentage.
P20 - Min et al. (2023)	Long-form generation mixes supported and unsupported content, so a single correctness label is uninformative.	Decompose generated text into atomic facts and verify each independently against a knowledge source; report the proportion supported.	Biography generation grounded in Wikipedia.	FactScore - fraction of atomic facts supported by the reference source.
P21 - Cao et al. (2026)	Hallucination datasets are expensive to annotate manually, which limits benchmark scale and coverage.	Automated pipeline that induces and collects factuality hallucinations from LLMs to construct labelled datasets without manual annotation.	Automatically generated hallucination datasets built from open-domain QA sources.	Hallucination detection accuracy; agreement with human labels.
P22 - Shuster et al. (2021)	Knowledge-grounded dialogue systems frequently state facts not present in the retrieved knowledge.	Compare retrieval-augmented dialogue architectures and knowledge-conditioning strategies against parametric baselines, with human evaluation of factual consistency.	Wizard of Wikipedia, CMU DoG and related knowledge-grounded dialogue corpora.	Human-judged factual consistency, knowledge F1, perplexity.
P23 - Asai et al. (2024)	Standard RAG retrieves a fixed number of passages regardless of need and never checks whether the output is actually supported.	Train the LM to emit reflection tokens that decide when to retrieve, and to critique relevance of retrieved passages and support for its own generations; segment-level selection at inference.	PopQA, TriviaQA, PubHealth, ARC-Challenge, ASQA, biography generation.	Accuracy, FactScore, citation precision and recall.
P24 - Yan et al. (2024)	RAG degrades badly when the retriever returns irrelevant documents, and standard pipelines have no recovery path.	Lightweight retrieval evaluator scores document relevance and triggers one of three actions - correct, incorrect or ambiguous - with web search as a fallback and decompose-then-recompose filtering of retrieved text.	PopQA, Biography, PubHealth, Arc-Challenge.	Accuracy, FactScore.

Paper	Key Findings	Advantages	Limitations	Future Scope
P13 - Huang et al. (2025)	Provides the now-standard factuality/faithfulness split; identifies retrieval augmentation as a principal mitigation while noting it introduces its own failure modes.	Comprehensive and current; explicitly connects hallucination causes to mitigation families.	Breadth over depth; being a survey it does not evaluate the compared methods under a common protocol.	Standardised benchmarks and mitigation that addresses causes rather than symptoms.
P14 - Zhang et al. (2023)	Argues hallucination is a systemic consequence of the training objective rather than an incidental defect, and that mitigation must span the whole pipeline.	Clear organisation and accessible synthesis of a fast-moving area.	Preprint; overlaps substantially with other contemporaneous surveys.	Causal analysis of hallucination and lifecycle-wide mitigation.
P15 - Xu et al. (2024)	Hallucination is an innate limitation of LLMs and cannot be fully removed by scaling, better data or guardrails; it can only be bounded in practice.	Sets a principled ceiling on what mitigation can achieve and reframes the objective as risk management rather than elimination.	Idealised assumptions; the practical gap between the theoretical bound and deployed error rates is not quantified.	Bounding achievable hallucination rates for realistic, restricted task distributions.
P16 - Kalai et al. (2025)	Hallucination persists because evaluation rewards guessing over saying 'I do not know'; changing benchmark scoring to reward calibrated abstention is proposed as a primary fix.	Shifts the problem from model internals to incentive design, which is directly actionable for evaluation practice.	Recent preprint; the proposed scoring reforms are not yet empirically validated at scale.	Benchmarks and training objectives that explicitly reward calibrated uncertainty.
P17 - Manakul et al. (2023)	Self-consistency across sampled generations detects hallucination effectively without any external resource, exploiting that fabricated content is less stable across samples.	Zero-resource and black-box compatible; strong practical baseline.	Requires several generations, so inference cost multiplies; confidently repeated falsehoods are consistent and therefore missed.	Cheaper consistency estimation and combination with retrieval-based verification.
P18 - Farquhar et al. (2024)	Entropy over semantic clusters detects confabulations substantially better than lexical uncertainty measures and generalises across datasets and models without task-specific supervision.	Principled, unsupervised, model-agnostic; published in a top general-science venue with strong empirical validation.	Requires multiple samples plus entailment computation, so it is costly; targets confabulation specifically, not consistent errors learned from training data.	Efficient approximations and integration into retrieval-triggering decisions.
P19 - Lin et al. (2022)	Larger models are often *less* truthful because they better model the human text distribution including its falsehoods - inverse scaling on truthfulness.	Demonstrated that scale alone does not fix factuality; became a standard evaluation.	Small and adversarial by construction, so absolute scores are not representative of general use; contamination risk given its age and popularity.	Truthfulness training objectives and evaluations resistant to memorisation.

Paper	Key Findings	Advantages	Limitations	Future Scope
P20 - Min et al. (2023)	Fine-grained atomic scoring reveals substantial unsupported content in outputs that read as fluent and coherent, and an automated estimator tracks human judgement closely.	Fine-grained, interpretable, and now a standard metric for long-form factuality including RAG output.	Depends on knowledge-source coverage; atomic-fact decomposition is itself model-generated and can err; primarily validated on biographies.	Extension to domains without a clean reference corpus.
P21 - Cao et al. (2026)	Automatically constructed hallucination corpora support detector training and evaluation at a scale manual annotation cannot reach, with quality sufficient for benchmarking.	Addresses the annotation bottleneck; recent IEEE journal publication with a reproducible pipeline.	Automatically induced hallucinations may not match the distribution of naturally occurring ones; label noise is inherited from the generating model.	Validating that induced hallucinations transfer to naturally occurring failure modes.
P22 - Shuster et al. (2021)	Retrieval augmentation substantially reduces hallucinated statements in conversation relative to parametric models, while maintaining conversational quality.	Among the first direct empirical demonstrations that retrieval reduces hallucination rather than merely improving accuracy.	Confined to dialogue; reduction is partial, and the system still hallucinates when retrieval returns weak evidence.	Grounding verification and handling retrieval failure explicitly.
P23 - Asai et al. (2024)	Self-reflective retrieval and self-critique outperform both standard RAG and much larger instruction-tuned baselines on factuality and citation accuracy.	Unifies adaptive retrieval and self-verification in one trained model; controllable at inference.	Requires training a specialised model with reflection-token supervision, so it does not apply to closed APIs; critique is itself model-generated and fallible.	Transferring reflective behaviour to black-box models via prompting.
P24 - Yan et al. (2024)	Explicitly assessing retrieval quality and correcting or replacing poor retrievals improves robustness substantially and is plug-and-play over existing RAG stacks.	Model-agnostic; directly targets retrieval failure, the dominant RAG error source; no generator retraining.	Adds an evaluator and possible web calls, increasing latency and cost; evaluator errors cascade; web fallback introduces an uncontrolled source.	Cheaper relevance estimation and principled fallback source selection.

Paper	Research Problem	Methodology	Dataset / Benchmark	Evaluation Metrics
P25 - Jiang et al. (2023)	Retrieving once at the start is insufficient for long-form generation, where information needs arise as text is produced.	FLARE: anticipate the upcoming sentence, and if it contains low-confidence tokens, use it as a query to retrieve before generating that sentence.	Multi-hop QA, commonsense reasoning, long-form QA and open-domain summarisation (2WikiMultiHopQA, StrategyQA, ASQA, WikiAsp).	Exact Match, F1, ROUGE, citation quality.
P26 - Trivedi et al. (2023)	Multi-step questions need evidence that only becomes identifiable after partial reasoning, which one-shot retrieval cannot supply.	IRCoT interleaves chain-of-thought reasoning with retrieval, using each generated reasoning step as the query for the next retrieval round.	HotpotQA, 2WikiMultiHopQA, MuSiQue, IIRC.	Retrieval recall, answer Exact Match and F1.
P27 - Dhuliawala et al. (2024)	Models assert unsupported facts confidently and do not check their own claims before answering.	Chain-of-Verification: draft a response, plan verification questions, answer each independently to avoid conditioning on the draft, then revise the response.	Wikidata list questions, QUEST, MultiSpanQA, longform biography generation.	Precision of listed facts, FactScore-style factuality.
P28 - Gao et al. (2023)	Editing a generated text for factuality while preserving its content requires attribution to real evidence.	RARR: research the generated text by issuing queries, retrieve evidence, then revise the output minimally to agree with the evidence while preserving the original intent.	Outputs over Natural Questions, StrategyQA and QReCC.	Attribution (AIS-style) and preservation of the original output.
P29 - Edge et al. (2024)	Standard RAG retrieves local text chunks and therefore cannot answer global, corpus-level questions such as 'what are the main themes'.	Build an entity knowledge graph from the corpus with an LLM, detect hierarchical communities, pre-generate community summaries, and answer global queries by map-reduce over those summaries.	Podcast transcripts and news article corpora; global sensemaking question sets.	LLM-judged comprehensiveness, diversity and empowerment; token cost.
P30 - Es et al. (2024)	RAG pipelines have separately failing retrieval and generation stages, but are usually evaluated only end-to-end and with human labels.	Reference-free framework computing faithfulness, answer relevance and context relevance using LLM-based judging of decomposed statements.	WikiEval and standard RAG evaluation settings.	Faithfulness, answer relevance, context relevance; agreement with human annotators.
P31 - Chen et al. (2024)	It is unclear which specific abilities RAG demands of an LLM and where the paradigm breaks.	Retrieval-Augmented Generation Benchmark (RGB) isolating four abilities - noise robustness, negative rejection, information integration and counterfactual robustness.	RGB, constructed from news with controlled noise and counterfactual injection, in English and Chinese.	Accuracy, rejection rate, error detection rate.
P32 - Niu et al. (2024)	Hallucination detection for RAG lacked a large word-level annotated corpus of genuinely retrieval-augmented outputs.	Corpus of nearly 18,000 naturally generated RAG responses with word-level hallucination annotation across QA, data-to-text and summarisation, plus a trained detector.	RAGTruth, built over Natural Questions, MARCO, Yelp and CNN/Daily Mail sources.	Span-level and response-level hallucination detection precision, recall and F1.
P33 - Zhang et al. (2025)	Mitigation techniques for retrieval-augmented LLMs are scattered and not organised by where in the pipeline they act.	Structured review decomposing RAG into retrieval-phase and generation-phase sub-tasks and mapping hallucination causes and mitigations onto each.	Review of the RAG hallucination literature.	Surveys the evaluation metrics used across the reviewed work.
P34 - Wolk (2025)	Which RAG variant should actually be deployed in a safety-critical setting, and at what latency and hallucination cost.	Controlled comparison of twelve RAG variants - dense, sparse, hybrid, graph-based, multimodal, self-reflective, adaptive and security-focused - on clinical vignettes, with on-premises deployment analysis.	250 de-identified patient vignettes with clinical reference material.	Precision@5, MRR, nDCG@10, hallucination rate, latency.
P35 - Tikka et al. (2026)	Whether laboratory hallucination-mitigation results hold up in real industrial deployments.	Applied study of hallucination mitigation across three distinct industrial use cases, reporting the techniques used and the operational outcomes.	Proprietary industrial datasets from the three deployment settings.	Task-specific accuracy and hallucination rates reported per use case.

Paper	Key Findings	Advantages	Limitations	Future Scope
P25 - Jiang et al. (2023)	Actively deciding when and what to retrieve during generation outperforms single-shot and fixed-interval retrieval on long-form knowledge-intensive tasks.	Uses generation confidence as the retrieval trigger, so retrieval happens where the model is actually uncertain.	Repeated retrieval raises latency; token probability is an imperfect proxy for factual uncertainty and needs likelihood access.	Better uncertainty signals and cost-aware retrieval scheduling.
P26 - Trivedi et al. (2023)	Interleaving retrieval with reasoning markedly improves both retrieval recall and answer accuracy on multi-hop questions relative to one-shot retrieval.	Simple, needs no training, and generalises across model sizes; a strong multi-hop baseline.	Multiple retrieval and generation rounds are expensive; an early reasoning error steers all subsequent retrieval wrongly.	Error recovery within the reasoning-retrieval loop and adaptive step budgets.
P27 - Dhuliawala et al. (2024)	Deliberate self-verification with independently answered verification questions reduces hallucination substantially, and independence from the draft is essential to the gain.	Prompt-level only, so it works on closed models with no training or retrieval infrastructure.	Several extra generation passes multiply cost; verification uses the same parametric knowledge, so shared misconceptions survive.	Combining verification questions with retrieval so checks consult external evidence.
P28 - Gao et al. (2023)	Post-hoc research-and-revise improves attribution substantially with limited change to the original text, and can be applied to any generator's output.	Fully post-hoc and model-agnostic; explicitly balances factuality against preservation.	Cannot fix a fundamentally wrong output, only locally repair it; dependent on retrieval quality; adds a full extra pipeline.	Deciding when revision is insufficient and regeneration is required.
P29 - Edge et al. (2024)	Graph-structured indexing answers corpus-level questions that chunk-based RAG cannot address, with large gains in comprehensiveness and diversity of answers.	Extends RAG from local lookup to global sensemaking; hierarchical summaries support multiple query granularities.	Index construction is expensive in LLM calls; evaluation relies on LLM judges rather than ground truth; graph extraction errors propagate into every downstream answer.	Cheaper graph construction and objective evaluation of global query answering.
P30 - Es et al. (2024)	Component-wise reference-free evaluation correlates well with human judgement and isolates whether a failure originated in retrieval or generation.	No gold answers required, so it can run continuously in production; widely adopted as the RAGAS library.	Uses an LLM as judge, inheriting that model's biases and cost; metric stability across judge models is not guaranteed.	Judge-independent metrics and validation across domains.
P31 - Chen et al. (2024)	Even strong LLMs degrade sharply with retrieval noise, rarely reject unanswerable queries, and frequently follow retrieved evidence that contradicts their correct parametric knowledge.	Diagnostic rather than aggregate; the four-ability decomposition is directly actionable for system design.	Constructed rather than naturally occurring noise; snapshot of models at time of writing.	Training explicitly for rejection and counterfactual robustness.

Paper	Key Findings	Advantages	Limitations	Future Scope
P32 - Niu et al. (2024)	A detector fine-tuned on RAGTruth rivals prompt-based detection with much stronger models, and hallucination persists in RAG output even when relevant evidence is retrieved.	The key supervised resource for RAG hallucination; word-level granularity; naturally occurring rather than induced errors.	Restricted to the tasks and generator models sampled; annotation of partial support is inherently subjective.	Broader task and language coverage; detectors that generalise across generators.
P33 - Zhang et al. (2025)	Hallucination causes are separable by pipeline stage - data source, query formulation, retriever and ranking on one side; grounding and decoding on the other - and mitigation should be matched to stage.	Directly organised around the RAG pipeline, which makes it practically usable; recent peer-reviewed journal synthesis.	Survey without unified empirical comparison; coverage of agentic RAG is limited.	Stage-aware benchmarks and combined multi-stage mitigation.
P34 - Wolk (2025)	Hybrid dense-sparse retrieval with cross-encoder reranking gave the best retrieval accuracy, while self-reflective RAG achieved the lowest hallucination rate; sparse retrieval was fastest but least accurate.	Rare like-for-like empirical comparison of many variants under one protocol; addresses privacy and deployment, not just accuracy.	Single clinical domain and a modest vignette set; results may not transfer to open-domain use.	Cross-domain replication of the comparison protocol.
P35 - Tikka et al. (2026)	Mitigation techniques transfer to industry but require domain-specific tuning, and residual hallucination remains material enough to require human oversight in production.	Deployment evidence rather than benchmark evidence; documents the practical gap between research and production.	Proprietary data limits reproducibility; small number of cases restricts generalisation.	Open industrial benchmarks and standardised operational reporting.

Reflection

Which methodology appears most frequently?

Dense passage retrieval feeding a generative reader is the base architecture in the large majority of the corpus, established by [P02] and [P01] and still the substrate that later work modifies rather than replaces. The most frequent *modification* is adding a decision point: whether to retrieve ([P10], [P25], [P23]), whether the retrieved evidence is usable ([P24]), or whether the generated claim is supported ([P23], [P27]). The trajectory is from a fixed pipeline toward a control loop.

Which dataset appears most frequently?

Natural Questions, followed by TriviaQA and PopQA. That concentration is itself a finding and a limitation of the field: these are predominantly English, entity-centric, Wikipedia-grounded factoid benchmarks. Conclusions drawn from them may not transfer to the specialist domains where hallucination is most consequential, which is precisely what [P34] and [P35] encounter when they move to clinical and industrial data.

Which evaluation metric appears most frequently?

Exact Match and token F1 dominate the earlier work, which is a poor fit for the problem because they measure answer correctness rather than whether the answer was supported by the retrieved evidence. The shift to grounding-aware metrics - FActScore [P20], RAGAS faithfulness [P30], span-level annotation [P32] - is one of the clearest signs of the field maturing, and it happens around 2023-24.

Phase 4 - Prompt Engineering Activity Log

Activity	Objective	Prompt Technique	Quality (1-5)	Improvements Made
Paper summary	Compress each paper into the eight matrix fields without losing the qualifiers that make a finding conditional.	Role prompting + structured output + explicit refusal clause	4	First version let the model infer datasets from context; it filled in plausible-but-wrong benchmarks for three papers. Adding NOT STATED and the no-inference rule eliminated this.
Dataset analysis	Determine which benchmarks actually recur across the corpus rather than which are most memorable.	Chain-of-Thought + explicit disambiguation constraint	5	Added the do-not-merge rule after the model collapsed NQ variants into one row, which inflated the apparent frequency of Natural Questions.
Method comparison	Compare mitigation architectures without producing an invalid cross-paper accuracy ranking.	Role prompting + negative constraint + CoT	5	The unconstrained version produced a confident leaderboard ranking Self-RAG against FiD on incomparable setups. Forbidding the ranking and requiring setup attribution was the single most valuable prompt change in the study.
Limitation extraction	Mine stated limitations from each paper, distinguishing author-acknowledged from reviewer-inferred.	Structured extraction + provenance separation	4	Early runs blended inferred limitations into the authors' own, which would have misrepresented the papers. The (a)/(b) split fixed it and made the inferred column defensible in its own right.
Gap analysis	Derive research gaps from limitation frequency rather than asking the model to invent them.	Evidence-grounded CoT with an explicit prohibition on ungrounded generation	5	This follows the manual's instruction not to ask AI to 'find research gaps'. Asked directly, the model returned generic gaps ('needs more multimodal work') unconnected to the corpus. Deriving from the frequency table produced gaps traceable to specific papers.
Draft sectioning	Produce section skeletons for the review paper without letting the model introduce unsourced claims.	Prompt chaining + closed-world constraint	4	The [GAP] marker was more useful than expected: it surfaced six places where the argument needed a citation I had not yet collected, rather than silently papering over them.

Prompts Used (verbatim)

Paper summary - Role prompting + structured output + explicit refusal clause

Act as a research analyst. Summarise the paper below into exactly these fields: research problem, methodology, dataset/benchmark, evaluation metrics, key findings, advantages, limitations, future scope. Rules: quote the paper's own words for any numeric result and give the section it came from. If a field is not stated in the paper, write NOT STATED - do not infer it from the abstract or from what similar papers do. Preserve every conditional ('on this dataset', 'for this model size').

Dataset analysis - Chain-of-Thought + explicit disambiguation constraint

From the extracted matrix rows below, list every dataset mentioned, with the paper IDs citing it. Do not merge datasets with similar names (Natural Questions and NQ-open are different; PopQA and EntityQuestions are different). Return a frequency table sorted by count, then state which three datasets dominate and what that concentration implies for the external validity of this literature.

Method comparison - Role prompting + negative constraint + CoT

Compare the systems below on: retrieval trigger, verification mechanism, training requirement, evaluation datasets. Do NOT produce a single ranked accuracy table. These systems use different base models, retrievers and splits, so a numeric ranking would compare experimental setups rather than methods. Where you cite a number, attach the base model and dataset it was measured on. State explicitly which comparisons are invalid and why.

Limitation extraction - Structured extraction + provenance separation

Extract every limitation from the paper below. Separate them into (a) limitations the authors state explicitly, quoting the sentence, and (b) limitations you infer that the authors do not acknowledge, each flagged INFERRED with your reasoning. Do not present (b) as if it were (a). If the paper acknowledges no limitations, say so - that is itself a finding.

Gap analysis - Evidence-grounded CoT with an explicit prohibition on ungrounded generation

Here is a frequency table of limitation categories across 35 papers, with the paper IDs in each category. For each category appearing in four or more papers, state: what specifically is missing, which papers evidence it, and why it matters practically. Ground every claim in the paper IDs given. Do NOT propose gaps that are not supported by this table, and do not suggest a gap merely because it sounds novel.

Draft sectioning - Prompt chaining + closed-world constraint

Draft the Comparative Analysis section using ONLY the matrix rows and comparison table supplied below. Every factual sentence must carry a paper ID. If a transition needs a claim not present in the supplied material, write [GAP: <what is needed>] instead of writing the claim. Do not add background you believe to be true but which is not in the input.

Reflection

Why was this prompting technique selected?

Each technique was matched to a distinct failure. Role prompting set the register and the standard of evidence for summarisation. Structured output with an explicit NOT STATED option prevented the model filling unstated fields with plausible defaults - the dominant failure in paper summarisation. Chain-of-Thought was used where the reasoning needed auditing, particularly dataset disambiguation. Negative constraints ('do NOT produce a ranked accuracy table') were the highest-value technique in the whole study, because the model's default behaviour was to produce exactly the invalid artefact I needed to avoid. Prompt chaining kept drafting closed-world, so each section could only use material established in an earlier step.

Did AI hallucinate?

Yes, in three distinguishable ways. Fabricated bibliographic records were the most frequent and the most dangerous: asked directly for citations, the model produced correctly formatted references to papers that do not exist, and to real papers with the wrong venue or year. This is why no citation in this review came from a model. Second, unstated experimental details were confabulated - benchmarks and metrics attributed to papers that never used them. Third, and subtlest, the model produced a confident cross-paper accuracy leaderboard that was internally coherent and methodologically invalid, because the systems compared used different base models and splits. The third kind is the hardest to catch, because nothing about it looks wrong.

How was the output verified?

Bibliographic data was never accepted from a model: all 45 records were resolved programmatically against CrossRef and the arXiv API by the submitted `fetch\_papers.py`, so title, author list, year and venue come from the publisher's record. Matrix content was checked against the paper's own abstract and results sections, with a NOT STATED convention rather than inference. Numeric claims carry a confidence flag in `comparison.csv`; anything marked `check` is listed in the workbook's verification appendix for re-reading before submission. Research gaps were derived by hand from the limitation frequency table rather than requested from a model, following the manual's explicit instruction.

Phase 5 - Research Evolution Analysis

Year	Major Contribution	Paper IDs	Impact
2017-2018	Transformer architecture and large-scale pre-training establish the parametric-knowledge paradigm; FEVER frames fact verification as a retrieval-plus-inference task.	Context for P09	Sets up the assumption that knowledge lives in weights - the assumption the rest of this decade spends undoing.
2019	LAMA probing shows LMs recall substantial relational knowledge but unevenly and with high prompt sensitivity.	P09	Establishes empirically that parametric memory is real but unreliable, motivating external retrieval.
2020	DPR replaces lexical matching with learned dense retrieval; REALM trains retrieval jointly during pre-training; RAG unifies a dense retriever with a generative reader end to end.	P01, P02, P03	The founding year of RAG. Knowledge becomes modular and swappable rather than frozen in weights.
2021	Fusion-in-Decoder scales evidence aggregation across many passages; retrieval augmentation is shown to reduce hallucination in dialogue; FAISS makes billion-scale vector search practical.	P04, P06, P22	First direct evidence that retrieval reduces hallucination specifically, plus the infrastructure that makes deployment feasible.
2022	RETRO demonstrates trillion-token retrieval lets small models match much larger ones; TruthfulQA shows scale can worsen truthfulness.	P05, P19	Decouples knowledge from parameter count and shows scaling alone does not solve factuality - inverse scaling on truthfulness.
2023	Retrieval becomes adaptive and iterative - FLARE retrieves on low confidence, IRCoT interleaves retrieval with reasoning; black-box methods (REPLUG, In-Context RALM) work without model access; SelfCheckGPT and FactScore make hallucination measurable; adaptive retrieval is justified by long-tail analysis.	P07, P10, P17, P20, P25, P26, P28	The pivot year: retrieval shifts from a fixed pre-processing step to a decision the model makes during generation, and hallucination becomes quantifiable.
2024	Self-reflection and correction enter the pipeline (Self-RAG, CRAG, Chain-of-Verification); structure enters the index (GraphRAG, RAPTOR); evaluation matures (RAGAS, RGB, RAGTruth); 'Lost in the Middle' and semantic entropy expose context and uncertainty limits.	P11, P18, P23, P24, P27, P29, P30, P31, P32	RAG stops being a pipeline and becomes a control loop with verification. Evaluation shifts from end-to-end accuracy to component-wise diagnosis.
2025	Theoretical limits are formalised (hallucination shown to be innate; incentive analysis of why models guess); agentic RAG surveys consolidate multi-step tool-using retrieval; peer-reviewed journal syntheses and controlled variant comparisons appear.	P13, P15, P16, P33, P34	The field matures from technique proliferation to consolidation, and accepts that elimination is not achievable - the goal becomes bounded, measurable residual risk.
2026	Industrial deployment studies and automated hallucination corpus construction move the work from benchmarks to production and from manual to scalable annotation.	P21, P35	Evidence base shifts from laboratory benchmarks to operational reporting, exposing the gap between benchmark and production performance.

What technological shifts occurred over the last decade?

Four, in sequence. Knowledge moved out of parameters and into an external index (2020: [P01], [P02], [P03]). Retrieval scaled, both in corpus size [P05] and in engineering feasibility [P06]. Retrieval became a decision rather than a fixed step, triggered by confidence or by a trained policy (2023: [P25], [P10]). Finally verification entered the loop, so the system began checking its own retrieved evidence and its own output (2024: [P23], [P24], [P27]). The underlying movement is from 'fetch then generate' to 'decide, fetch, check, and abstain if unsupported'.

Which techniques became obsolete?

Purely lexical retrieval as a standalone method - though BM25 survives as a component of hybrid retrieval, which [P34] finds still gives the best retrieval accuracy when fused with dense retrieval. Single-shot always-on retrieval is superseded for anything multi-step. Relying on scale alone for factuality was effectively ended by [P19], which showed larger models can be less truthful. And Exact Match as a sufficient evaluation of a knowledge-intensive system is now clearly inadequate, since it cannot distinguish a grounded answer from a lucky one.

Which techniques became dominant?

Hybrid dense-sparse retrieval with cross-encoder reranking as the retrieval layer; adaptive triggering rather than always-on retrieval; self-critique or external verification before the answer is returned; and component-wise, reference-free evaluation [P30] instead of end-to-end accuracy alone. Structured indexing ([P29], [R42]) is dominant specifically for corpus-level questions that chunk retrieval cannot serve.

Phase 6 - Domain-Specific Comparison Study

Why this table is architectural rather than numeric

The manual's AI/ML matrix asks for Accuracy, Precision, Recall and F1. Those columns are deliberately not populated here. The systems compared use different base models, retrievers, corpora and evaluation splits, so a single accuracy ranking would compare experimental setups rather than methods and would favour whichever paper used the strongest base model. The comparison is therefore made on dimensions that are genuinely commensurable, with headline results reported only alongside the setup that produced them. This is argued in Section IV of the review paper.

Paper	Model / System	Retrieval Trigger	Verification	Training Req.	Dataset	Reported Headline Result	Limitation
P01	RAG (Sequence/Token)	Single-shot, always	None	Generator + retriever fine-tuned	NQ, TriviaQA, WebQ, FEVER	NQ Exact Match ~44	No support checking; single retrieval round
P02	DPR	Single-shot, always	None	Dual-encoder supervised	NQ, TriviaQA, WebQ, TREC	Large top-20 retrieval gain over BM25	Needs labelled pairs; weak out of domain
P03	REALM	Single-shot, always	None	Joint retriever + MLM pre-training	NQ, WebQ, CuratedTrec	Outperforms comparable parametric baselines on open QA	Very expensive pre-training
P04	Fusion-in-Decoder	Single-shot, always	None	Reader fine-tuned	NQ, TriviaQA	Accuracy scales with passage count	Cost linear in passages; no sufficiency signal
P05	RETRO	Chunk-level, always	None	Pre-trained with chunked cross-attention	MassiveText, Pile, Wikitext103	Small model matches far larger parametric models	Frozen retriever; leakage control required
P07	In-Context RALM	Fixed stride	None	None (frozen LM)	WikiText-103, Pile	Substantial perplexity reduction on a frozen LM	Context-window bound; noise hurts
P08	REPLUG	Single-shot, always	None	Retriever tuned to LM likelihood	Pile, NQ, TriviaQA, MMLU	Gains on a black-box LM without model access	Needs token likelihoods
P10	Adaptive retrieval by popularity	Adaptive (popularity threshold)	None	None	PopQA, EntityQuestions	Retrieval helps the long tail, hurts head entities	Popularity is a crude confidence proxy
P22	Retrieval-augmented dialogue	Per-turn, always	None	Fine-tuned	Wizard of Wikipedia	Marked drop in hallucinated statements vs parametric	Partial reduction only; dialogue-specific
P23	Self-RAG	Adaptive (reflection token)	Self-critique of relevance and support	Trained with reflection tokens	PopQA, PubHealth, ARC, ASQA, Bio	Beats standard RAG and larger instruction-tuned baselines on factuality and citation	Needs a specially trained model; critique is fallible
P24	CRAG	Single-shot + corrective	Retrieval evaluator + web fallback	Lightweight evaluator	PopQA, Bio, PubHealth, ARC	Improved robustness under poor retrieval; plug-and-play	Extra latency; evaluator errors cascade
P25	FLARE	Adaptive (low token confidence)	None explicit	None	2WikiMultihop, StrategyQA, ASQA, WikiAsp	Beats single-shot and fixed-interval retrieval on long-form tasks	Latency; needs likelihood access
P26	IRCoT	Iterative (per reasoning step)	None	None	HotpotQA, 2WikiMultihop, MuSiQue, IIRC	Large recall and EM gains on multi-hop	Expensive; early errors misdirect retrieval
P27	Chain-of-Verification	No retrieval (parametric)	Independent verification questions	None	Wikidata lists, QUEST, MultiSpanQA	Substantial hallucination reduction via self-verification	Shares the model's own misconceptions
P28	RARR	Post-hoc research	Attribution-based revision	None	NQ, StrategyQA, QReCC outputs	Improved attribution with high preservation of original text	Repairs locally; cannot fix wholly wrong output
P29	GraphRAG	Global map-reduce over summaries	None	None (LLM index build)	Podcast and news corpora	Answers corpus-level questions chunk RAG cannot	Costly index; LLM-judge evaluation
P34	12 RAG variants compared	Varies by variant	Varies by variant	Varies	250 clinical vignettes	Self-reflective RAG lowest hallucination; hybrid retrieval best P@5	Single domain; modest sample

Phase 7 - Limitation Mining

Paper	Limitation 1	Limitation 2	Limitation 3
P01 - Lewis et al.	Verification absence	Retrieval quality	Context utilisation
P02 - Karpukhin et al.	Generalization	Dataset issues	Retrieval quality
P03 - Guu et al.	Computational cost	Generalization	Retrieval quality
P04 - Izacard et al.	Computational cost	Verification absence	Context utilisation
P05 - Borgeaud et al.	Retrieval quality	Computational cost	Reproducibility
P06 - Johnson et al.	Retrieval quality	Generalization	Computational cost
P07 - Ram et al.	Context utilisation	Retrieval quality	Verification absence
P08 - Shi et al.	Generalization	Retrieval quality	Reproducibility
P09 - Petroni et al.	Evaluation validity	Dataset issues	Generalization
P10 - Mallen et al.	Evaluation validity	Generalization	Retrieval quality
P11 - Liu et al.	Generalization	Evaluation validity	Context utilisation
P12 - Ji et al.	Generalization	Evaluation validity	Dataset issues
P13 - Huang et al.	Evaluation validity	Generalization	Dataset issues
P14 - Zhang et al.	Evaluation validity	Reproducibility	Generalization
P15 - Xu et al.	Evaluation validity	Generalization	Reproducibility
P16 - Kalai et al.	Evaluation validity	Reproducibility	Generalization
P17 - Manakul et al.	Computational cost	Verification absence	Evaluation validity
P18 - Farquhar et al.	Computational cost	Generalization	Evaluation validity
P19 - Lin et al.	Dataset issues	Evaluation validity	Generalization
P20 - Min et al.	Dataset issues	Generalization	Evaluation validity
P21 - Cao et al.	Dataset issues	Evaluation validity	Generalization
P22 - Shuster et al.	Generalization	Verification absence	Retrieval quality
P23 - Asai et al.	Computational cost	Generalization	Verification absence
P24 - Yan et al.	Computational cost	Retrieval quality	Reproducibility
P25 - Jiang et al.	Computational cost	Generalization	Verification absence
P26 - Trivedi et al.	Computational cost	Retrieval quality	Verification absence
P27 - Dhuliawala et al.	Computational cost	Verification absence	Generalization
P28 - Gao et al.	Retrieval quality	Computational cost	Verification absence
P29 - Edge et al.	Computational cost	Evaluation validity	Retrieval quality
P30 - Es et al.	Evaluation validity	Computational cost	Generalization
P31 - Chen et al.	Dataset issues	Generalization	Evaluation validity
P32 - Niu et al.	Generalization	Dataset issues	Evaluation validity
P33 - Zhang et al.	Evaluation validity	Generalization	Reproducibility
P34 - Wolk	Generalization	Dataset issues	Reproducibility
P35 - Tikka et al.	Reproducibility	Generalization	Dataset issues

Limitation Frequency

Limitation Category	Frequency	Share (%)
Generalization	26	24.8
Evaluation validity	18	17.1
Computational cost	14	13.3
Retrieval quality	13	12.4
Dataset issues	11	10.5
Verification absence	10	9.5
Reproducibility	9	8.6
Context utilisation	4	3.8

Three limitations were coded per paper, giving 105 codings across 35 studies. This distribution is the evidence base for the research gaps in Phase 8 - the gaps are derived from it rather than requested from a model.

Phase 8 - Research Gap Discovery

Method note

The manual states that students must not ask AI to 'find research gaps'. That instruction was followed. Each gap below was derived from the Phase 7 frequency table by taking categories appearing in four or more papers and identifying what is specifically missing. For comparison, a model asked directly for gaps returned generic suggestions ('more multimodal work is needed') that were untraceable to any paper in the corpus; those were discarded.

Gap ID	Research Gap	Supporting Papers	Evidence	Why It Matters
G1	No standard way to decide when retrieval should fire	P10; P23; P25; P31	P10 shows always-on retrieval degrades head-entity answers while helping the long tail. P25 triggers on token confidence and P23 on trained reflection tokens - two incompatible signals, never compared under one protocol. P31 shows models rarely reject unanswerable queries, so they cannot recognise when retrieval has failed.	Retrieval triggering is currently a per-system heuristic. Every deployment re-derives it, and no benchmark isolates trigger quality from retrieval or generation quality, so improvements cannot be attributed.
G2	Grounding is assumed rather than verified at generation time	P01; P04; P22; P32	P01 and P04 contain no mechanism to check that output is entailed by retrieved passages. P22 reduces but does not eliminate hallucination. P32 demonstrates hallucination persists in RAG output even when relevant evidence was successfully retrieved - the failure is in grounding, not retrieval.	The central assumption of RAG - that retrieved evidence constrains generation - is empirically false at the margin. Without per-claim entailment checking, a RAG system cannot distinguish a supported answer from a fluent unsupported one.
G3	Verification cost is never reported alongside accuracy	P17; P18; P23; P25; P26; P27	Computational cost is the most frequent limitation in this corpus. P17 and P18 need multiple samples; P25 and P26 retrieve repeatedly; P27 adds verification passes. None of these report latency or token cost against the hallucination reduction achieved.	Practitioners cannot choose a technique without a cost-benefit figure. A method that halves hallucination at five times the latency is not comparable to one that reduces it by a third at 1.2x, yet the literature presents them as if they were.
G4	Cross-paper numeric comparison is invalid but routinely performed	P30; P31; P32; P34	Reported systems use different base models, retrievers, corpora and splits. P34 is the only study in this corpus comparing many variants under one protocol, and it covers a single clinical domain. P30 provides component-wise metrics but with an LLM judge whose stability across models is unvalidated.	Without a shared protocol the field cannot establish which mitigation actually works. Apparent progress may reflect differences in base model strength rather than in method.
G5	Evaluation increasingly depends on LLM judges whose reliability is unestablished	P29; P30; P19; P20	P29 evaluates global query answering with LLM-judged comprehensiveness and diversity. P30's faithfulness metric is LLM-computed. P20's atomic-fact decomposition is model-generated. P19 uses a fine-tuned judge model.	Hallucination evaluation is becoming circular - models are increasingly judged by models with correlated failure modes. If judge and generator share a misconception, the metric certifies the error.
G6	Long-tail and low-resource domains remain unaddressed	P10; P02; P34; P35	P10 identifies the long tail as where retrieval matters most, yet P02-style retrievers need supervised in-domain pairs and degrade out of domain. P34 and P35 report that domain-specific tuning is required in clinical and industrial settings.	The settings where hallucination is most consequential - specialist medical, legal, industrial - are exactly those with the least training data for retrievers and the least benchmark coverage.
G7	Theoretical impossibility results are disconnected from practical error budgets	P15; P16; P35	P15 proves hallucination cannot be eliminated; P16 argues evaluation incentives sustain it. Neither connects to an achievable residual rate for a bounded task. P35 reports residual hallucination in production requiring human oversight, but without reference to any theoretical floor.	Engineering needs a target. Without a link between the impossibility results and attainable rates on restricted domains, deployment decisions about acceptable risk have no principled basis.

Phase 9 - Future Research Directions

Gap	Proposed Research Direction	Expected Impact	Feasibility
G1	A unified retrieval-triggering benchmark that holds retriever and generator fixed and varies only the trigger policy (always-on, popularity, token confidence, semantic entropy, trained reflection).	Makes trigger quality independently measurable and comparable for the first time.	High - composable from existing public datasets and released trigger implementations.
G2	Per-claim entailment gating: decompose the draft into atomic claims, check each against retrieved passages with an NLI model, and force abstention or re-retrieval on unsupported claims before the answer is returned.	Converts grounding from an assumption into an enforced constraint, and yields a per-claim support score for the user.	High - combines mature atomic decomposition and NLI components.
G3	Mandatory cost-quality reporting: a standard triple of hallucination rate, added latency and token overhead for every proposed mitigation.	Enables rational technique selection and exposes methods whose gains are simply bought with compute.	High - a reporting convention, not a technical advance.
G4	A frozen common-protocol harness fixing base model, retriever, corpus and splits, against which any mitigation can be dropped in.	Separates method contribution from base-model strength; makes progress claims falsifiable.	Medium - requires community coordination and sustained maintenance.
G5	Judge-independence auditing: evaluate every LLM-judged metric across several judge families and report inter-judge agreement as a validity statistic.	Quantifies how much of a reported gain is an artefact of the judge, and detects correlated generator-judge failure.	High - a protocol addition to existing frameworks.
G6	Retriever domain adaptation from unlabelled corpora plus small expert-verified seed sets, evaluated specifically on long-tail entities.	Extends RAG reliability to the specialist domains where hallucination carries the highest cost.	Medium - unsupervised adaptation is unsolved but actively progressing.
G7	Empirical error-budget curves: measure achievable residual hallucination as a function of domain restriction and abstention rate, linking impossibility results to deployment targets.	Gives engineers a defensible acceptable-risk figure rather than an aspiration of zero.	Medium - needs careful task-distribution definition.

Novel Research Ideas

Idea 1: Entailment-Gated RAG (EG-RAG)

Question	Response
Problem Solved	RAG systems emit unsupported claims even when correct evidence was retrieved (P32). No production pipeline enforces claim-level grounding before returning an answer.
Supporting Papers	P32 (hallucination persists despite good retrieval); P20 (atomic decomposition); P23 (self-critique); P27 (independent verification)
Why Novel?	Existing work either critiques with the generating model itself (P23, P27) - inheriting its misconceptions - or scores factuality after the fact (P20). EG-RAG puts an independent, non-generative NLI gate inside the response path, making abstention the default when entailment fails rather than an option the model may decline to take.
Expected Impact	A per-claim support score returned with every answer, and a tunable precision-abstention trade-off that a deployment can set by risk tolerance.

Idea 2: Entropy-Triggered Adaptive Retrieval (ETAR)

Question	Response
Problem Solved	Retrieval triggers use token probability (P25) or trained reflection tokens (P23). Token probability confuses lexical with semantic uncertainty; reflection tokens need a specially trained model and so exclude closed APIs.
Supporting Papers	P18 (semantic entropy separates meaning-level from wording-level uncertainty); P25 (confidence-triggered retrieval); P10 (retrieval should be selective); P23
Why Novel?	Semantic entropy has been used only for post-hoc detection (P18). Using it as the retrieval trigger inverts it from a diagnostic into a controller, and unlike reflection tokens it needs no model training, so it works on black-box models.
Expected Impact	Retrieval fires when the model is uncertain about meaning rather than phrasing - fewer wasted retrievals on head entities and fewer misses on the long tail.

Idea 3: Adversarial Judge Auditing for RAG metrics (AJA)

Question	Response
Problem Solved	RAG evaluation increasingly relies on LLM judges (P29, P30) whose reliability is unestablished, risking circular evaluation where judge and generator share failure modes.
Supporting Papers	P30 (LLM-judged faithfulness); P29 (LLM-judged global answers); P19 (fine-tuned judge); P16 (evaluation incentives shape model behaviour)

Question	Response
Why Novel?	Current practice validates a judge once against human labels on one dataset. AJA instead constructs adversarial cases targeting known shared misconceptions between judge and generator, and reports a judge-independence statistic as a required component of any evaluation claim.
Expected Impact	Turns judge reliability from an assumption into a reported measurement, and detects the specific failure where a metric certifies an error because judge and generator agree.

Phase 10 - Review Paper Drafting

Section	AI Role Used	Prompting Technique	Human Modifications Made	Final Status
Abstract	Academic Writer	Role prompting	Rewrote to lead with the corpus size and the specific finding rather than generic framing; added the 2019-2026 span.	Completed
Introduction	Research Analyst	Chain-of-Thought	Reordered from technique-first to problem-first; added the parametric-memory motivation from P09 and P10 which the draft omitted.	Completed
Research Methodology	Research Analyst	Structured output	Added the exact CrossRef/arXiv verification procedure and inclusion criteria; the AI draft described a generic PRISMA-style process that did not match what was actually done.	Completed
Literature Review	Research Reviewer	Prompt chaining	Regrouped from chronological to thematic; removed four claims that no collected paper supported.	Completed
Comparative Analysis	Technical Analyst	ReAct + negative constraint	Replaced the AI's accuracy leaderboard with an architectural comparison and added the explicit statement that cross-paper numeric ranking is invalid here.	Completed
Research Trends	Research Analyst	CoT over the evolution table	Added the 2023 inflection argument (retrieval becomes a decision) which the draft presented only as a list of papers.	Completed
Research Gaps	Research Mentor	CoT + role prompting	Derived entirely from the limitation frequency table by hand; the AI's direct gap suggestions were discarded as ungrounded.	Completed
Future Directions	Research Consultant	Role prompting	Added feasibility ratings and removed two proposals that restated existing published work as novel.	Completed
Conclusion	Academic Writer	Prompt chaining	Rewritten in own words; cut the AI's closing flourish about 'transforming the future of AI' and replaced it with the specific residual-risk argument.	Completed

Phase 11 - Humanization and Originality Improvement

Section	AI Draft Words	Final Words	Delta	Major Changes Made
Abstract	210	178	-32	Removed three hedging phrases ('it is important to note', 'plays a crucial role'); replaced generic claims with the corpus figures.
Introduction	690	612	-78	Cut a paragraph of AI-typical scene-setting; added the specific P09/P10 parametric-memory evidence and two citations.
Research Methodology	380	470	+90	Expanded - the AI version was too thin. Added the actual verification pipeline, inclusion/exclusion criteria and the publisher distribution.
Literature Review	1180	1075	-105	Removed four unsupported claims; regrouped thematically; added explicit limitations to each subsection.
Comparative Analysis	820	910	+90	Deleted the invalid accuracy leaderboard entirely; wrote the architectural comparison and the methodological caveat by hand.
Research Trends	520	486	-34	Replaced list-like prose with the inflection-point argument; removed repeated transition phrasing.
Research Gaps	610	742	+132	Almost entirely rewritten. AI-suggested gaps discarded; replaced with gaps derived from the limitation frequency table, each tied to paper IDs.
Future Directions	540	598	+58	Added feasibility assessments and the three novel proposals with their novelty justifications.
Conclusion	340	302	-38	Rewritten in own language; removed grandiose closing and stated the residual-risk position instead.

AI draft total 5290 words; final total 5373 words (+1.6%). The net figure understates the rewriting: Research Gaps and Comparative Analysis were substantially replaced rather than edited, so word count is close while content is not.

What domain knowledge did you manually add?

The methodological argument that cross-paper numeric comparison is invalid here, which no draft produced and which now shapes the whole Comparative Analysis section. The 2023 inflection reading of the evolution timeline - that the field's turning point was retrieval becoming a decision rather than a step. The connection between [P32]'s finding and the paradigm's core assumption. And the observation that the benchmark concentration on Wikipedia-grounded factoid QA limits the external validity of nearly every result reported here.

What content was removed because it sounded AI-generated?

Formulaic openers ('In today's rapidly evolving landscape'), throat-clearing transitions ('It is important to note that'), tricolon padding where three near-synonyms did the work of one word, and uniformly hedged conclusions that committed to nothing.

The clearest tell was paragraph-level symmetry: every subsection arriving at the same length with the same three-part shape. Real analysis is uneven, because some papers deserve more space than others.

Which sections required major rewriting?

Research Gaps was rewritten almost completely - the AI's directly suggested gaps were generic and untraceable to the corpus, so they were discarded and replaced with gaps derived from the limitation frequency table, each tied to specific paper IDs. Comparative Analysis required deleting the accuracy leaderboard entirely and rebuilding the section around architectural dimensions. Research Methodology had to be expanded rather than cut, because the draft described a generic systematic-review process instead of the verification pipeline actually used.

Phase 12 - AI Detection and Plagiarism Validation

These figures must be yours

Similarity, AI-detection, grammar and readability scores can only be produced by running the finished paper through the tools your institution uses (Turnitin, GPTZero, Grammarly or equivalent). They are left blank deliberately - a fabricated originality score would be the exact failure this case study is about. Run the scans on output/REVIEW_PAPER_IEEE.docx and record the results here.

Parameter	Required	Result	Tool used	Date
Similarity Index	< 10%			
AI Detection Score	< 10%			
Grammar Score	> 90%			
Readability Score	Good			

Phase 13 - Final Review Paper Submission

The complete review paper is submitted as a separate document, **output/REVIEW_PAPER_IEEE.docx**, in two-column IEEE format with numbered citations in order of first appearance.

#	Section
1	Title
2	Abstract
3	Keywords
4	Introduction
5	Research Methodology
6	Literature Review
7	Comparative Analysis
8	Research Trends
9	Research Gaps
10	Future Research Directions
11	Conclusion
12	References

Appendix A - Faculty Evaluation Rubric

Criteria	Marks
Topic Selection & Scope	5
Paper Collection (30+)	10
Literature Review Master Matrix	15
Prompt Engineering Portfolio	10
Research Evolution Analysis	10
Comparative Study	10
Limitation Mining	10
Research Gap Identification	10
Future Scope & Novel Ideas	10
Humanization & Originality	5
Final Review Paper Quality	5
Total	100

Appendix B - Figures to Re-check Before Submission

Every numeric result quoted in this submission carries a confidence flag. The rows below are figures recalled from reading rather than re-read from the paper during compilation. Open each paper, confirm the figure, and set figure_confidence to 'high' in comparison.csv.

Paper	System	Figure to confirm	Checked
P01	RAG (Sequence/Token)	NQ Exact Match ~44	[]

Appendix C - Reproducibility

File	Purpose
fetch_papers.py	Resolves all 45 records against CrossRef and the arXiv API
seed_data.py	Writes the Phase 3-11 analysis tables
content.py	All prose and the full review paper text
build.py	Renders the workbook and the IEEE paper; rennumbers citations
data/papers.csv	Verified bibliographic repository
data/matrix.csv	Phase 3 master matrix, 35 papers x 8 fields